Health-a-thon 2026

FINAL SUBMISSION

GlucoFlow

Source-linked coordination for inpatient diabetes care

TEAM NAME **BetaRevive**

Team Overview & Background

- BetaRevive is developing GlucoFlow for the operational work around inpatient diabetes care.
- Engineering contributions: a web workspace, authenticated API, source-evidence pipeline and bounded Hermes integration.
- The supplied clinical workflow notes shaped the focus on handover gaps and inconsistent records.

Problem Statement and Supporting Evidence

- A specimen can be collected without a recorded laboratory receipt. A validated result can still await acknowledgement.
- At shift change, staff must reconstruct what happened, who owns the next step and which source confirms completion.
- The supplied notes identify cross-department communication and recording discrepancies. These are design inputs, not measured prevalence or clinical outcome evidence.

Target Users and Use Case

- Primary users: inpatient doctors, nursing staff and laboratory teams.
- Initial setting: a synthetic DKA episode, including documented renal-team handover context. HHS is an optional transfer-handover scenario.
- The job: inspect outstanding work, check the source, approve a bounded task and prepare the receiving team's handover. Clinical decisions remain with the treating team.

Solution Overview, Key Features & Impact

- Ask about outstanding specimen or result handoffs.
- Open the source event and inspect the responsible owner.
- Approve a time-bounded operational task.
- Record the matching simulated event and verify completion.
- Review a handover with unresolved work and evidence.
- Distinctive capability: an approved procedure can propose work for another encounter, with new source checks and a separate approval.
- Expected benefit: less manual chasing. No measured reduction is claimed.

Technology, Tools and AI Approach

- Interface: React/TypeScript chat, OpenUI evidence views and Sarvam speech.
- Agent: native Hermes with constrained care tools. The recorded live-provider deployment used gpt-6-luna.
- Authority: Fastify validates references and permissions; PostgreSQL persists tasks, approvals and audit history.
- Hosting: Vercel frontend, Azure API/runtime and Supabase services.

Synthetic HIS/LIS feeds. Live hospital interfaces and ABDM integration are not implemented.

Build Sprint Journey and Progress

- Initial foundation: report intake, source review and approved history.
- Care extension: encounter events, operational tasks and source-confirmed completion.
- Agent integration: Hermes tools, reviewed handovers and versioned procedures.
- 7 Nov refinement: assistive-only scope, compact chat, direct speech controls and simpler navigation.
- Final demo pass: visible New chat, DKA focus and optional HHS template.
- Open work: hospital integration, full device verification and a clinician pilot.

Milestone order reflects repository evidence. A verified week-by-week partner record was not supplied.

Evaluation and User Validation

- Automated evidence: 282 application tests passed in the 7 Nov navigation release. Latest focused care suite: 15/15 passed. These are separate runs.
- Hosted checks: clinic and patient entry, new simulation, HHS encounter persistence and 390 px layout.
- Team feedback: removed duplicate screens and large panels; added a visible New chat control.
- AI limits: a successful provider turn is not a clinical accuracy benchmark. Repetitive help replies and response latency need further work.

Next validation: clinician usability, full browser regression, device speech and measured latency/cost.

Safety and Responsible AI

- Only fictional data in the evaluation environment. Demo accounts are shared.
- No diagnosis, prescribing, risk scoring or treatment thresholds.
- Server-side permissions and exact source references limit agent authority.
- Human approval gates operational tasks and record publication. Corrections retain history.
- Unavailable feeds remain unknown. Model calls have budgets and deadlines.
- English/Hindi support with visibly partial language previews. Device speech and broader language review remain open.

Deployment, Cost and Sustainability Plan

- Pilot proposal: one ward, named owners, agreed HIS/LIS event imports and a privacy review. No EHR replacement or live ABHA claim.
- One-time planning budget: ₹40,000, assuming 40 setup/training hours at ₹1,000/hour. Hospital connector work is excluded.
- Monthly planning budget: ₹6,000/facility: ₹3,000 hosting, ₹2,000 AI/speech and ₹1,000 support.
- Usage assumption: 2,000 interactions/month at a ₹1 variable-cost allowance. Budget hypotheses, not measured bills or vendor quotes.
- Validate willingness to pay before offering a facility subscription.

Next Steps

- Days 1–30: agree event definitions, review privacy requirements and measure the existing baseline.

Days 31–60: run a supervised shadow pilot with human review of every operational action.

- Days 61–90: compare median verified collection-to-receipt lag and missing-receipt rate. Track review burden and incorrect completion separately.
- Expansion depends on reliable event data and staff usability evidence. No hospital adoption or clinical benefit is claimed.

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Thank you

BetaRevive

https://glucoflow.vercel.app/
Select Explore the clinic, then Create workspace
Shared synthetic demo. No manual credentials needed.

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